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# Dynamic fracture mechanics

L. B. FREUND

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*DYNAMIC FRACTURE MECHANICS*

# *Dynamic Fracture Mechanics*

*L. B. FREUND*

*Brown University*

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# PREFACE

This book is an outgrowth of my involvement in the field of dynamic fracture mechanics over a period of nearly twenty years. This subbranch of fracture mechanics has been wonderfully rich in scope and diversity, attracting the attention of both researchers and practitioners with backgrounds in the mechanics of solids, applied mathematics, structural engineering, materials science, and earth science. A wide range of analytical, experimental, and computational methods have been brought to bear on the area. Overall, the field of dynamic fracture is highly interdisciplinary, it provides a wealth of challenging fundamental issues for study, and new results have the potential for immediate practical application. In my view, this combination of characteristics accounts for its continued vitality.

I have written this book in an effort to summarize the current state of the mechanics of dynamic fracture. The emphasis is on fundamental concepts, the development of mathematical models of phenomena which are dominated by mechanical features, and the analysis of these models. Mathematical problems which are representative of the problem classes that comprise the area are stated formally, and they are also described in common language in an effort to make their features clear. These problems are solved using mathematical methods that are developed to the degree required to make the presentation more or less self-contained. Experimental and computational approaches have been of central importance in this field, and relevant results are cited in the course of discussion. The extraordinary contributions of the few individuals who pioneered the area of dynamic fracture mechanics occupy prominent positions in this discussion. One hope in preparing this book is that people with new perspectives will be attracted to the field, which continues to provide

fascinating and technically important challenges. Perhaps the book can serve as a guide to further development of the area.

The reader is assumed to be familiar with concepts of continuum mechanics and methods of applied mathematics to the level normally provided through the first year of study in a graduate program in solid mechanics in the United States. A brief summary of relevant results is included in the first chapter in order to establish notation and to provide a common source for reference in later chapters. Some background in equilibrium fracture mechanics would be helpful, of course, but none is presumed. In terms of graduate instruction in fracture mechanics, the book could serve as a text for a course devoted to dynamic fracture mechanics or, for a more general course, as a supplement to other books which provide broader coverage of the whole of fracture mechanics. The overall organization of the book is evident from the chapter titles. The brief overview included in Chapter 1 can serve as a guide to those readers interested in using the book as a source of reference for specific results.

The bibliography is an important part of the book. In view of contemporary publication practices, the compilation of an all-inclusive bibliography in any technical area is an impossible task. Nonetheless, the bibliography is intended to be comprehensive in the sense that it includes entries which describe research results on essentially all aspects of dynamic fracture. With only a few exceptions, the entries are either articles published in the open literature or relevant textbooks and monographs. Thus, most of the references should be available in a reasonably complete technical library. All references cited in the text are included in the bibliography, and many additional references are included as well. Some judgment was required in the selection of references for citation in the text, and I have done my best to accurately identify sources for key steps in the evolution of ideas.

I am indebted to a number of colleagues who read drafts of various chapters of the book. Those who offered suggestions and encouragement in this way are John Hutchinson, Fred Nilsson, Ares Rosakis, and John Willis. A special thanks goes to Jim Rice, Editor of this series, who generously read a draft of the entire manuscript. It has been among my greatest fortunes to be a member of the Solid Mechanics Group at Brown University, which has provided an intellectually stimulating and most congenial environment over the years. I am especially grateful to my colleagues Rod Clifton, Alan Needleman, Michael Ortiz, Fong Shih, and Jerry Weiner for their willingness to

read and discuss some of the material that is included here. My own views on the mechanics of dynamic fracture and its fundamental precepts have been formed over a long period of time through interactions with people far too numerous to mention individually. This group includes the many colleagues and students with whom I have collaborated and written joint papers; they are identified in the bibliography. It also includes those who, through questions and discussions, showed me where my own understanding of certain points had been incomplete.

I also thank Peter-John Leone, Earth Sciences Editor, Rhona Johnson, Manuscript Development Editor, and Louise Calabro Gruendel, Production Editor, of Cambridge University Press in New York for the efficient way in which they have managed the preparation of the book and for their sensitivity in dealing with my concerns on the matter. It has been a pleasure to be involved in long-term programs of research on fracture mechanics funded by the Office of Naval Research and by the National Science Foundation. These programs, and the collaborations that they have fostered, have been invaluable.

Finally, I thank my wife Colleen and our sons, Jon, Jeff, and Steve, who enthusiastically adopted the mission of writing this book as their own. Their interest and unwavering devotion have lightened the task immensely.

*L. B. Freund*

# LIST OF SYMBOLS

Mathematical symbols and functions are defined the first time that they are used in the book. Brief definitions of the most frequently used symbols and functions are listed below. Some symbols necessarily have different definitions in different sections. In such cases, definitions are stated locally and are used consistently within sections.

| | |
|---|---|
| $a$ | Inverse dilatational wave speed $c_d^{-1}$; half length of a Griffith crack |
| $A_I(v)$ | Universal function of crack tip speed for mode I deformation (similar for modes II and III) |
| $b$ | Inverse shear wave speed $c_s^{-1}$; a material parameter |
| $c$ | Inverse Rayleigh wave speed $c_R^{-1}$; an elastic wave speed |
| $c_d$ | Elastic dilatational wave speed |
| $c_0$ | Speed of longitudinal waves in an elastic bar |
| $c_R$ | Elastic Rayleigh surface wave speed |
| $c_s$ | Elastic shear wave speed |
| $C_I$ | Dimensionless factor for the mode I asymptotic stress field (similar for modes II and III) |
| $C_{ijkl}$ | Components of the elastic stiffness tensor |
| $d$ | Inverse crack tip speed $v^{-1}$ |
| $d_{ij}$ | Components of the symmetric part of the velocity gradient tensor |
| $D(v)$ | The quantity $4\alpha_d\alpha_s - (1+\alpha_s^2)^2$; a function of crack tip speed |
| $E$ | Young's elastic modulus |
| $E_R$ | Far-field radiated energy |
| $F(\Gamma)$ | Energy flux through contour $\Gamma$ |
| $G$ | Energy release rate; dynamic energy release rate |

| | |
|---|---|
| $G_a$ | Crack arrest fracture energy |
| $G_c$ | Critical value of energy release rate |
| $h$ | Thickness of a beam; width of a strip |
| $h_i(x,t)$ | Components of the dynamic weight function |
| $H(t)$ | Unit step function |
| $J_N(\Gamma, s)$ | Path independent integral of Laplace transformed fields |
| $k(v)$ | Universal function of crack tip speed for elastic crack growth in mode I |
| $k_{II}(v)$ | Universal function of crack tip speed for elastic crack growth in mode II (similar for mode III) |
| $K_I$ | Elastic stress intensity factor for mode I (similar for modes II and III) |
| $K_{Ia}$ | Value of stress intensity factor at crack arrest; crack arrest toughness (similar for modes II and III) |
| $K_{Iappl}$ | Remotely applied stress intensity factor |
| $K_{Ic}$ | Value of stress intensity factor at fracture initiation; fracture toughness (similar for modes II and III) |
| $K_{Id}$ | Value of stress intensity factor during crack growth; dynamic fracture toughness (similar for modes II and III) |
| $l, l(t)$ | Crack length; amount of crack growth |
| $\dot{l}$ | Instantaneous crack tip speed |
| $l^{\pm}$ | Limit as position $x = l$ is approached through values of $x$ greater than $l$ (+) or less than $l$ (−) |
| $l_c$ | Critical crack length |
| $l_0$ | Initial crack length |
| $m$ | Normalized crack tip speed $v/c_s$ or $v/c_R$ |
| $M_{ijkl}$ | Components of the elastic compliance tensor |
| $n$ | Crack tip bluntness parameter; a material parameter |
| $n_i$ | Components of unit vector; normal to surface or curve |
| $o(f(x))$ | Asymptotically dominated by $f$ as $x \to$ a limit point |
| $O(f(x))$ | Asymptotically proportional to $f$ as $x \to$ a limit point |
| $p(x)$ | Pressure distribution |
| $p^*$ | Magnitude of a concentrated normal force |
| $P(\zeta)$ | Amplitude of $\Phi$ |
| $P_c$ | Dimensionless combination of material parameters for a strain-rate-dependent elastic-plastic solid |
| $q^*$ | Magnitude of a concentrated shear force |
| $Q(\zeta)$ | Amplitude of $\Psi$ |

| | |
|---|---|
| $r$ | Polar coordinate |
| $r_d \exp(i\theta_d)$ | Polar form of the complex variable $x + i\alpha_d y$ |
| $r_p$ | Plastic zone size |
| $r_s \exp(i\theta_s)$ | Polar form of the complex variable $x + i\alpha_s y$ |
| $R$ | Region in space; region in a plane |
| $R(\zeta)$ | Rayleigh wave function |
| $s$ | Laplace transform parameter; a real variable |
| $s_{ij}$ | Components of the deviatoric stress tensor |
| $S_\pm(\zeta)$ | Factors of the Rayleigh wave function that are nonzero and analytic in overlapping half planes |
| $t$ | Time coordinate |
| $T$ | Kinetic energy density |
| $T_{tot}$ | Total kinetic energy of a body |
| $\mathbf{T}, T_i$ | Traction vector; components of traction vector |
| $u_i$ | Components of particle displacement vector ($i = 1, 2, 3$ or $i = x, y, z$) |
| $\dot{u}_i$ | Components of particle velocity vector |
| $\ddot{u}_i$ | Components of particle acceleration vector |
| $u_y^L(x,t)$ | Normal surface displacement for Lamb's problem |
| $u_-(x,t)$ | Displacement distribution for $x < 0$ |
| $U$ | Stress work density; elastic strain energy density |
| $U_{tot}$ | Total stress work; total strain energy |
| $v$ | Crack tip speed |
| $w$ | Displacement for antiplane shear deformation |
| $x_i$ | Rectangular coordinates ($i = 1, 2, 3$) |
| $x, y, z$ | Rectangular coordinates |
| $\alpha(\zeta)$ | The function $(a^2 - \zeta^2)^{1/2}$ |
| $\alpha_d$, $\alpha_s$ | The quantities $\sqrt{1 - v^2/c_d^2}$ and $\sqrt{1 - v^2/c_s^2}$ |
| $\beta(\zeta)$ | The function $(b^2 - \zeta^2)^{1/2}$ |
| $\gamma$ | A material parameter |
| $\dot{\gamma}_o$ | Viscosity parameter of a rate dependent plastic material |
| $\Gamma$ | Crack tip contour; specific fracture energy |
| $\Gamma(\cdot)$ | Gamma (factorial) function |
| $\Gamma_c, \Gamma_m, \Gamma_o$ | Constant values of specific fracture energy |
| $\delta_t$ | Crack tip opening displacement |
| $\delta(t)$ | Dirac delta function |
| $\Delta$ | Amplitude of an elastic dislocation |
| $\epsilon$ | A small real parameter |
| $\epsilon_{ij}$ | Components of the small strain tensor |

| | |
|---|---|
| $\epsilon_{ij}^{e}$ | Components of the elastic strain tensor |
| $\epsilon_{ij}^{p}$ | Components of the plastic strain tensor |
| $\zeta$ | Complex variable; Laplace transform parameter |
| $\eta$ | Rectangular coordinate |
| $\theta$ | Polar coordinate |
| $\kappa$ | Parameter determined by the behavior of the Rayleigh wave function as $\lvert\zeta\rvert \to \infty$ |
| $\lambda$ | Lamé elastic constant |
| $\Lambda$ | Length of cohesive zone |
| $\mu$ | Lamé elastic constant; elastic shear modulus |
| $\nu$ | Poisson's ratio |
| $\boldsymbol{\nu}, \nu_i$ | Unit vector normal to a surface or curve |
| $\xi$ | Rectangular coordinate |
| $\rho$ | Material mass density |
| $\sigma$ | Mean stress; effective stress |
| $\sigma(x)$ | Normal traction within a cohesive zone |
| $\sigma_{ij}$ | Components of the stress tensor |
| $\sigma_\infty$ | Amplitude of remotely applied tension |
| $\sigma_o$ | Tensile flow stress of an ideally plastic material |
| $\sigma_+(x,t)$ | Normal traction distribution for $x > 0$ |
| $\sigma^*$ | Magnitude of applied normal traction |
| $\Sigma_{ij}^{I}$ | Angular variation of asymptotic crack tip stress field for mode I (similar for modes II and III) |
| $\tau_\infty$ | Amplitude of remotely applied shear traction |
| $\tau_o$ | Shear flow stress of an ideally plastic material |
| $\tau_+(x,t)$ | Shear traction distribution on $x > 0$ |
| $\tau^*$ | Magnitude of applied shear traction |
| $\phi$ | Lamé scalar displacement potential function |
| $\Phi$ | Double Laplace transform of $\phi$ |
| $\boldsymbol{\psi}, \psi_i$ | Lamé vector displacement potential function |
| $\psi$ | Magnitude of $\boldsymbol{\psi}$ for plane deformation; local shear angle in plastically deforming region |
| $\Psi$ | Double Laplace transform of $\psi$ |
| $\Omega$ | Potential energy |
| $\Omega_S$ | Potential energy increase due to creation of free surface |
| $(cc.ss.nn)$ | Equation number $nn$ in Section $ss$ of Chapter $cc$ |
| $(c.s.n)_m$ | The $m$th equation in a group of equations identified by the single number $(c.s.n)$ |

# 1

# BACKGROUND AND OVERVIEW

## 1.1 Introduction

The field of fracture mechanics is concerned with the quantitative description of the mechanical state of a deformable body containing a crack or cracks, with a view toward characterizing and measuring the resistance of materials to crack growth. The process of describing the mechanical state of a particular system is tantamount to devising a mathematical model of it, and then drawing inferences from the model by applying methods of mathematical or numerical analysis. The mathematical model typically consists of an idealized description of the geometrical configuration of the deformable body, an empirical relationship between internal stress and deformation, and the pertinent balance laws of physics dealing with mechanical quantities. For a given physical system, modeling can usually be done at different levels of sophistication and detail. For example, a particular material may be idealized as being elastic for some purposes but elastic-plastic for other purposes, or a particular body may be idealized as a one-dimensional structure in one case but as a three-dimensional structure in another case. It should be noted that the results of most significance for the field have not always been derived from the most sophisticated and detailed models.

A question of central importance in the development of a fracture mechanics theory is the following. Is there any particular feature of the mechanical state of a cracked solid that can be interpreted as a

"driving force" acting on the crack, that is, an effect that is correlated with a tendency for the crack to extend? A viewpoint that underlies this important concept is that of the crack as an entity which itself behaves according to a "law" of mechanics expressed in terms of a relationship between driving force and motion. The modeling phase provides the language for considering the strength of real materials, and it concludes with hypotheses on the behavior of materials. It is only through observation of the fracturing of materials that the hypotheses can be verified or refuted. This synergistic process has led to standard methods of practice whereby fracture mechanics is used routinely in materials selection and engineering design.

### *1.1.1 Inertial effects in fracture mechanics*

Dynamic fracture mechanics is the subfield of fracture mechanics concerned with fracture phenomena for which the role of material inertia becomes significant. Phenomena for which strain rate dependent material properties have a significant effect are also typically included. Inertial effects can arise either from rapidly applied loading on a cracked solid or from rapid crack propagation. In the case of rapid loading, the influence of the loads is transferred to the crack by means of stress waves through the material. To determine whether or not a crack will advance due to the stress wave loading, it is necessary to determine the transient driving force acting on the crack. In the case of rapid crack propagation, material particles on opposite crack faces displace with respect to each other once the crack edge has passed. The inertial resistance to this motion can also influence the driving force, and it must be taken into account in a complete description of the process. There is also a connection between the details of rapid crack motion and the stress wave field radiated from a moving crack that is important in seismology, as well as in some material testing techniques.

Progress toward understanding dynamic fracture phenomena has been impeded by several complicating features. The inherent time dependence of a dynamic fracture process results in mathematical models that are more complex than the equivalent equilibrium models for the same configuration and material class. From the experimental point of view, the time dependence requires that many accurate sequential measurements of quantities of interest must be made in an extremely short time period in a way that does not interfere with the process being observed. In the case of crack growth, the regions of the

boundary of the body over which certain conditions must be enforced in a mathematical model change with time. From the experimental point of view, this feature implies that the place where quantities of interest must be measured varies, usually in a nonuniform way, during the process.

The question of whether or not inertial effects are significant in any given fracture situation depends on the loading conditions, the material characteristics, and the geometrical configuration of the body. Circumstances under which they are indeed significant cannot be specified unambiguously, but some guidelines are evident. For example, suppose that there is a characteristic time associated with the applied loading, say a load maximum divided by the rate of load increase. Unless this time is large compared to the time required for a stress wave to travel at a characteristic wave speed of the material over a representative length of the body, say the crack length or the distance from the crack edge to the loaded boundary, it can be expected that inertial effects will be significant. In the case of crack growth, inertial effects will probably be important if the speed of the crack tip or edge is a significant fraction of the lowest characteristic wave speed of the material. If the boundaries of the body must be taken into account, then the potential for the body to function as a waveguide must be considered. The group velocity of guided waves is frequently much less than the characteristic bulk wave speeds of the material. For such cases, inertial effects could be important for crack speeds that are much less than the bulk wave speeds but comparable to relevant group velocities.

### *1.1.2 Historical origins*

Empirical studies concerned with the bursting of military cannon or with the impact loading of industrial machines were carried out in the 19th century. For example, an interesting early series of observations on "new" industrial iron-based materials is reported by Kirkaldy (1863). However, no single scientific discovery or other event can be identified as the stimulus for launching dynamic fracture mechanics as an area of research. Instead, the area as it is summarized in this book is the conjunction of numerous paths of investigation that were motivated by pressing practical needs in engineering design and material selection, by scientific curiosity about earthquakes and other natural phenomena, by challenges of classes of mathematical problems, and by advances in experimental and observational techniques.

The area has been under continuous development as a subfield of the engineering science of fracture mechanics since the 1940s and as a topical area in geophysics since the late 1960s. The view in the early days of fracture mechanics was that once a crack in a structure began to grow the structure had failed (Tipper 1962). The rapid crack growth phase of failure was viewed as interesting but not particularly important. Beginning about 1970, however, the importance of understanding crack propagation and arrest was recognized in both engineering and earth sciences, and significant progress has been made since then.

A few of the early contributions to the area that provided ideas of lasting significance are cited here. In some cases, these ideas were developed in connection with fracture under equilibrium conditions, but they have been important in the area of dynamic fracture as well. Among these is the work of Griffith (1920), which is commonly acknowledged as representing the start of equilibrium fracture mechanics as a quantitative science of material behavior. In considering an ideally brittle elastic body containing a crack, he recognized that the macroscopic potential energy of the system, consisting of the internal stored elastic energy and the external potential energy of the applied loads, varied with the size of the crack. He also recognized that extension of the crack resulted in the creation of new crack surface, and he postulated that a certain amount of work per unit area of crack surface must be expended at a microscopic level to create that area. In this context, the term "microscopic" implies that this work is not included in a continuum description of the process. It is common to characterize this work per unit area by assuming that a certain force-displacement relationship governs the reversible interaction of atoms or molecules across the fracture plane. The area under the force-displacement relationship from equilibrium to full separation, averaged over the fracture surface, is assumed to represent this work of separation. This is one way to define the *surface energy* of the material. Griffith simply included this work as an additional potential energy of the system, and then invoked the equilibrium principle of minimum potential energy for conservative systems. That is, he considered the system to be in equilibrium with a particular fixed loading and a particular crack length. He postulated that the crack was at a *critical state of incipient growth* if the reduction in macroscopic potential energy associated with a small virtual crack advance from that state was equal to the microscopic

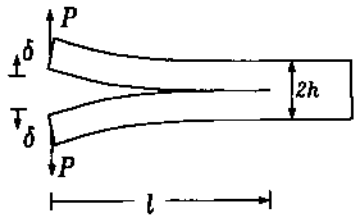


Figure 1.1. A crack of length $l$ in a rectangular strip. Forces $P$ act to separate the crack faces and the displacement of each load point is $\delta$.

work of separation for the crack surface area created by the virtual crack advance. A particular attraction of Griffith's energy fracture condition is that it obviates the need to examine the actual fracture process at the crack tip in detail.

Griffith's original work was based on the plane elasticity solution for a crack of finite length in a body subjected to a uniform remote tension in a direction normal to the crack plane. The concept is illustrated here by means of a simple example based on elementary elastic beam theory. Consider the split rectangular strip shown in Figure 1.1 which is of unit thickness in the direction perpendicular to the plane of the figure. Suppose that the slit or crack of length $l$ is opened symmetrically by imposing either forces or displacements at the ends of the arms as shown. In either case, the end force per unit thickness is denoted by $P$ and the displacement of the load point of each arm from the undeflected equilibrium position is denoted by $\delta$. Suppose that the isotropic material is elastic with Young's modulus $E$ and that the deformation can be approximated by assuming that each arm deforms as a Bernoulli-Euler beam of length $l$ cantilevered at the crack tip end. The relationship between the end force per unit thickness and the end displacement is then

$$P = \frac{Eh^3}{4l^3}\delta \tag{1.1.1}$$

and the total stored elastic energy per unit thickness in both arms is

$$\delta P = \frac{Eh^3}{4l^3}\delta^2 = \frac{4l^3}{Eh^3}P^2 . \tag{1.1.2}$$

The work of separation per unit area of surface created is denoted by $\gamma$.

Consider first the case of imposed end displacement $\delta^*$. What is the critical crack size for this fixed end displacement according to the Griffith theory? In this case, there is no energy exchange between the body and its surroundings during the virtual crack extension, and the macroscopic potential energy per unit thickness $\Omega$ is the stored elastic energy $\Omega = Eh^3\delta^{*2}/4l^3$. The total work-of-fracture per unit thickness is $\gamma$ times the crack length for both faces, or $\Omega_S = 2\gamma l$. The total work-of-fracture per unit thickness could equally well be written as $2\gamma$ times the amount of crack extension from some arbitrary initial crack length, a work measure that would differ from $\Omega_S$ by a constant. This constant is arbitrary, and it has no significance in considering crack advance. The total potential energy is $\Omega + \Omega_S$. According to the Griffith postulate, the critical crack length for incipient growth is determined by the condition

$$\frac{\partial}{\partial l}(\Omega + \Omega_S) = 0 \quad \Rightarrow \quad l_c = \left[\frac{3Eh^3\delta^{*2}}{8\gamma}\right]^{1/4}. \tag{1.1.3}$$

The individual potential energies vary with crack length as shown schematically in Figure 1.2. Note that

$$\frac{\partial^2}{\partial l^2}(\Omega + \Omega_S) > 0 \tag{1.1.4}$$

at the critical crack length. This means that the system is stable in the sense of mechanical equilibrium. If the crack length is at the critical value and $\delta^*$ is slowly increased to a larger value, then the crack will slowly advance under equilibrium conditions to a new length that is related to the increased $\delta^*$ through $(1.1.3)_2$. A useful interpretation of $(1.1.3)_1$ is that $-\partial\Omega/\partial l$ is the feature of the mechanical state that can be identified as a crack driving force. The driving force is work-conjugate to crack position $l$, and the fracture mechanics balance law governing crack behavior is $-\partial\Omega/\partial l = 2\gamma$.

Consider next the case of imposed end force $P^*$. What is the critical crack size for this end condition? The stored elastic energy is again given by (1.1.2). The external potential energy of the loading on both arms is $-2\delta P^* = -8l^3P^{*2}/Eh^3$, so that the macroscopic

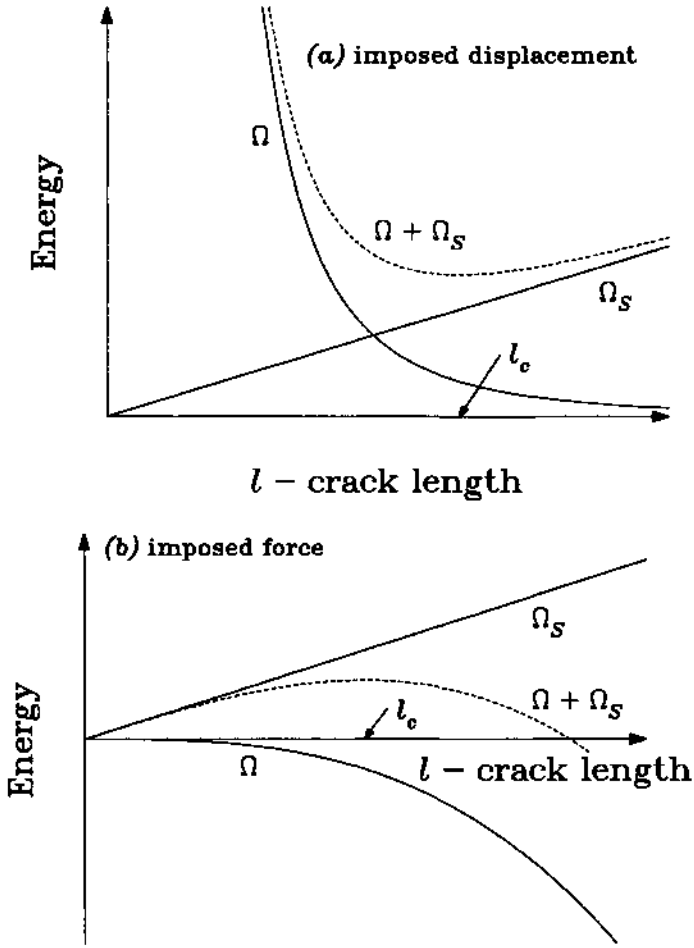


Figure 1.2. The variation of energy with crack length for the split rectangular strip in Figure 1.1 for (a) imposed displacement or (b) imposed force. The variation of the continuum potential energy plus the surface energy is shown by the dashed line.

potential energy is $\Omega = -4l^3P^{*2}/Eh^3$. Imposition of the Griffith condition in this case yields

$$\frac{\partial}{\partial l}(\Omega + \Omega_S) = 0 \quad \Rightarrow \quad l_c = \left[\frac{Eh^3\gamma}{6P^{*2}}\right]^{1/2} . \tag{1.1.5}$$

Again, the way in which the individual potential energies vary with crack size is illustrated schematically in Figure 1.2. If $\delta^*$, $P^*$, and

$l_c$ are related as in (1.1.1) then the critical lengths in (1.1.3b) and (1.1.5b) are identical. However, there is a fundamental difference between the two cases. In the latter case,

$$\frac{\partial^2}{\partial l^2}(\Omega + \Omega_S) < 0 \tag{1.1.6}$$

at the critical crack length. This means that the system is unstable in an equilibrium sense. When $l = l_c$ for a given $P^*$, the state is an equilibrium state of incipient crack growth. However, if $P^*$ is increased by any amount from this state, then it is no longer possible to find an equilibrium configuration. Instead, the crack begins to grow rapidly and inertial resistance to motion is called into play to ensure balance of momentum. This idea is fully developed in Chapter 7 in discussing dynamic crack growth.

A theoretical framework for including inertial effects during the rapid crack growth phase was first proposed by Mott (1948), who adopted the features of Griffith's analysis as a point of departure. He recognized that inertial resistance of the material to crack opening could become significant at high crack speeds. To estimate the crack speed for a particular loading system, he assumed that the crack growth process was steady state, that is, time independent as seen by an observer moving at speed $v$ with the crack tip. Under these conditions, he obtained an estimate for the total kinetic energy of the system $T_{tot}$ in the form of $v^2$ times a function of crack length $l$. He then argued that the total energy of the system $\Omega + \Omega_S + T$ is constant, so that

$$\frac{\partial}{\partial l}(\Omega + T_{tot} + \Omega_S) = 0\,. \tag{1.1.7}$$

This condition provides a relationship involving the loading parameters, the crack length $l$, and the crack speed $v$. In implementing this condition for the Griffith plane elasticity crack model, Mott assumed that $\Omega$ and $T$ could be calculated on the basis of the equilibrium field.

The idea underlying (1.1.7) is correct if the terms are strictly interpreted. In fact, it is a special case for steady state crack growth of the general energy-rate balance: Rate of work of applied loads = rate of increase of stored elastic energy + rate of increase of kinetic energy − rate of energy loss at the crack tip. However, later work has shown that the assumptions made by Mott in his original analysis of the dynamic Griffith crack are not valid. Consequently, the conclusions

inferred from this model are generally not valid. For example, this analysis has often been cited as establishing that the maximum crack speed in an ideal brittle material is some fixed fraction, typically given as about one-half, of the shear wave speed of the material. If the model is analyzed more thoroughly, it is found that this is not the case. Indeed, the result that the theoretical limiting speed of a tensile crack must be the Rayleigh wave speed, as established in Chapter 7, was anticipated by Stroh (1957) on the basis of an intuitive argument. Nonetheless, Mott's basic idea of energy balance during rapid crack growth was very important to the development of the subject. A number of energy concepts pertinent to dynamic fracture are discussed in Chapter 5, and the crack tip equation of motion of the type originally sought by Mott is presented in Chapter 7.

Another important theoretical idea for applying work and energy methods to fracture dynamics was provided by Irwin (1948). He was concerned with cleavage fracture of structural steels, a cleavage process that is invariably accompanied by some amount of plastic flow of the material adjacent to the fracture path. Irwin adopted Mott's postulate of stationary total energy, including a work of fracture, denoted above by $\Omega_S$. However, Irwin proposed that this term $\Omega_S$ may be approximately represented as the sum of two terms, one proportional to the area of fracture surface and the other proportional to the volume of material affected by plastic deformation. Within the context of the Griffith theory outlined above, this assumption is invoked by writing $\Omega_S = 2(\gamma_s + \gamma_p)l$ when the thickness of the plastically deformed layer adjacent to the crack faces is small. Thus, $\gamma_s$ represents the surface energy of the material associated with cleavage fracture and $\gamma_p$ represents the plastic work dissipated in the surrounding material per unit crack surface area created. Irwin's idea therefore extended the range of applicability of the energy balance fracture theory to include materials that undergo some plastic deformation during crack growth. This particular extension of Griffith's theory was also proposed for fracture initiation and crack propagation in steels by Orowan (1955). Early experiments aimed at measuring $\gamma_p$ for cleavage of metal single crystals and polycrystals were carried out by Hall (1953). He showed that $\gamma_p \gg \gamma_s$ for metals, particularly for polycrystals. Furthermore, both Hall and Irwin predicted strong connections between the fracture behavior of polycrystalline metals and their microstructures. An extensive survey of early work on the

connection between metallurgical properties and brittle fracture of welded steel plates was given by Wells (1961) and Tipper (1962).

The continuum field approach to fracture of solids was launched with the introduction of the *elastic stress intensity factor*, usually denoted by $K$, as a crack tip field characterizing parameter by Irwin (1957). This idea provided an alternate framework for discussing the strength of cracked solids of nominally elastic material. Irwin proposed that a crack will begin to grow in a cracked body with limited plastic deformation when $K$ is increased to a value called the *fracture toughness* of the material. The equivalence of the Irwin stress intensity factor criterion and the Griffith energy criterion for onset of growth of a tensile crack in a two-dimensional body of nominally elastic material under plane stress conditions was demonstrated by Irwin (1957, 1960), who showed that

$$-\frac{\partial \Omega}{\partial l} = \frac{K^2}{E}, \tag{1.1.8}$$

where $E$ is Young's modulus of the material. The quantity on the left side of (1.1.8) is usually denoted by $G$ and it is called the *energy release rate*. Much of the analysis of this book is devoted to determining the stress intensity factor or the energy release rate for various conditions. These concepts are developed more thoroughly in a way relevant to dynamic fracture mechanics in the chapters to follow.

Another idea that has been important in the evolution of fracture mechanics concerns the size scale over which different phenomena dominate a fracture process. The idea is implicit in Irwin's stress intensity factor concept and it is a central feature of the crack tip cohesive zone model introduced by Barenblatt (1959a). Consider a planar crack in a body subjected to tensile loading normal to the plane of the crack. Suppose that the material response is linearly elastic, except for a region adjacent to the crack edge where the response departs from linearity. The source of nonlinearity can be plastic deformation, diffuse microcracking, nonlinear interatomic forces, or some other physical mechanism. The crack tip region is said to be *autonomous* at fracture initiation or during crack growth if the following two conditions are met: (i) the extent of the region of nonlinearity from the crack edge is very small compared to all other length dimensions of the body and loading system, and (ii) the mechanical state within this end region at incipient growth or during growth is independent of loading

and geometrical configuration. For the particular case of an elastic-plastic material, the property of autonomy implies that the crack tip plastic zone is completely surrounded by an elastic stress intensity factor field and that the state within the plastic zone is determined by the level of stress intensity of the surrounding field. This situation, which was termed *small-scale yielding* by Rice (1968), is the situation in which the stress intensity factor is a useful fracture characterizing parameter. Furthermore, this close connection between effects at adjacent size scales provided the basis for development of an asymptotic formulation of nonlinear crack tip field problems in general. In its simplest form, the asymptotic formulation can be summarized in the following way. Suppose that, at one scale of observation, the mechanical state near a crack edge is dominated by an asymptotic *near crack tip solution* with some characterizing parameter. Then, to study the phenomenon at a physical size scale that is an order of magnitude smaller, a boundary value problem is formulated with a condition that its solution must match the aforementioned asymptotic solution *in the far-field.* It is essentially this asymptotic formulation that underlies the focus on semi-infinite cracks and near tip fields in this monograph, as well as in fundamental fracture mechanics studies in general. The central idea of autonomy has been extended to more general material classes than those mentioned in these introductory remarks.

A good deal of very important experimental research on dynamic fracture of materials was carried out in parallel with the development of dynamic fracture mechanics concepts for analyzing cracked solids. These experiments were often done on specimens of glass or other very brittle materials. The results of these experiments, in the form of both qualitative observations and quantitative data, have been extremely important in the development of dynamic fracture mechanics. Indeed, a number of specific observations on the behavior of growing cracks in glass were made in the 1940s and 1950s for which generally accepted explanations based on physical principles are still incomplete. For example, quantitative modeling as a basis for understanding terminal crack speed in glass or dynamic crack bifurcation under symmetric tensile loading is still in a preliminary stage. In addition to the importance of the experimental results obtained in the early stages of dynamic fracture research, the significance of the experimental techniques developed in the course of this work must also be recognized. The full-field optical methods of observation and special techniques

of high-speed photography have been particularly notable in dynamic fracture research.

The research group led by Schardin played a major role in the development of optical methods and high-speed photography. Much of this work is reviewed by Schardin (1959). For example, in 1937 it was found that running cracks in glass seem to have a maximum speed that depends on the material, but not on loading or geometrical configuration. This phenomenon was also observed for fracture of glass by Edgerton and Barstow (1941). The multiple spark camera, now known as the Cranz-Schardin camera in its most common configuration, was developed to obtain a sequence of photographs of a fracturing specimen in a very short time interval. It was also shown that stress waves obliquely incident on the crack path could cause the path to gradually curve toward the local direction of maximum tensile stress. It was observed that there was no delay in the change in crack growth direction due to incident stress waves, suggesting that the crack tip itself does not have an effective inertia (cf. Section 7.4). This phenomenon was exploited in the 1950s by Kerkhof, who superimposed ultrasonic waves on a crack growing rapidly in a brittle body. Because the frequency of the ultrasonic waves is known, the small undulations produced on the fracture surface due to the interaction of the crack with the ultrasonic waves provided a continuous record of crack tip speed (Kerkhof 1970, 1973).

The optical shadow spot method was also introduced through the work of this group (Manogg 1966). This technique made it possible to infer directly a value of the instantaneous crack tip stress intensity factor and the crack tip position. When used in conjunction with the multiple spark camera, a record of the stress intensity factor and crack motion could be obtained for a crack growth process. A multiple spark arrangement was also used by Wells and Post (1958) to determine the isochromatic fringe pattern representing the near crack tip stress field during rapid crack growth in a photoelastic brittle plastic. A review of early work on the formation of fractures in brittle materials by means of stress waves is given by Kolsky (1953). Modern surveys of experimental results in dynamic fracture of brittle materials are given by Dally (1987) for photoelastic methods and by Kalthoff (1987) for shadow spot methods.

## 1.2 Continuum mechanics

In this section, some notational conventions are established and basic equations of continuum mechanics are presented in order to provide a framework for discussing dynamic fracture mechanics. Field equations and integral relations are stated without derivation; the underlying developments can be found in the references cited. In presenting these results, it is tacitly assumed that the mathematical operations implied by the expressions can indeed be carried out. Whether or not this is the case in the study of any particular model must be determined in the course of analysis.

### *1.2.1 Notation*

Consider a three-dimensional Euclidean point space in which the behavior of a deformable body is to be described. A rectangular coordinate frame is introduced, consisting of a set of mutually orthogonal unit base vectors $\mathbf{e}_1$, $\mathbf{e}_2$, $\mathbf{e}_3$. The position vector $\mathbf{p}$ of a point in the space is

$$\mathbf{p} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + x_3\mathbf{e}_3 = \sum_{i=1}^{3} x_i\mathbf{e}_i = x_i\mathbf{e}_i\,, \tag{1.2.1}$$

where the summation over the repeated index implied by the last entry in the continued equality is adopted as a convention. The quantities $x_1, x_2, x_3$ are the rectangular coordinates of the point. Vector and tensor valued quantities are represented by boldface type $\mathbf{v}$ throughout the book, and components are represented by means of subscripts.

Suppose that $\mathbf{v}$ and $\mathbf{T}$ are a vector field and a second-order tensor field, respectively, over the space. At each point, the vector is an element of a vector space associated with the point space, and the second-order tensor is a linear transformation of elements of the vector space into possibly another vector space or, as in this case, the same one. The rectangular components of these fields at any point are

$$v_i = \mathbf{v}\cdot\mathbf{e}_i \quad \text{and} \quad T_{ij} = \mathbf{e}_i\cdot\mathbf{T}\cdot\mathbf{e}_j\,, \tag{1.2.2}$$

where the dot denotes the binary *inner product* operator. This notation is commonly called the Gibbs dyadic notation (Malvern 1969). The inner product of two vectors $\mathbf{v}$ and $\mathbf{u}$ is

$$\mathbf{u}\cdot\mathbf{v} = \mathbf{v}\cdot\mathbf{u} = u_iv_i\,. \tag{1.2.3}$$

Likewise, the inner product from the left (right) of vector $\mathbf{v}$ with the tensor $\mathbf{T}$ is $\mathbf{v}\cdot\mathbf{T} = v_i T_{ij}\mathbf{e}_j$ ($\mathbf{T}\cdot\mathbf{v} = \mathbf{e}_i T_{ij} v_j$). Inner products involving tensors of second- or higher order are not commutative unless the tensors possess certain symmetry properties. The cross product of two vectors $\mathbf{u}$ and $\mathbf{v}$ is a vector with components

$$(\mathbf{u}\times\mathbf{v})_i = -(\mathbf{v}\times\mathbf{u})_i = e_{ijk}u_j v_k\,, \tag{1.2.4}$$

where $e_{ijk}$ is the alternating symbol with values

$$e_{ijk} = \begin{cases} +1 & \text{if } ijk \text{ is an even permutation of 123,} \\ -1 & \text{if } ijk \text{ is an odd permutation of 123,} \\ 0 & \text{otherwise.} \end{cases} \tag{1.2.5}$$

The array of components of the second-order unit tensor $\mathbf{I}$ in the rectangular frame is the Kronecker delta

$$\delta_{ij} = \begin{cases} 1 & \text{for } i = j, \\ 0 & \text{for } i \neq j. \end{cases} \tag{1.2.6}$$

Thus, for any vector $\mathbf{v}$, $\mathbf{v} = \mathbf{I}\cdot\mathbf{v}$ or $v_i = \delta_{ij}v_j$. The trace of a second-order tensor $\mathbf{T}$ is

$$\mathrm{tr}(\mathbf{T}) = \mathbf{e}_i\cdot\mathbf{T}\cdot\mathbf{e}_i = T_{ii}\,. \tag{1.2.7}$$

Let $p$ represent the set of coordinates $x_1, x_2, x_3$. The gradient of a scalar field $f(p)$, denoted by $\nabla f$, is the vector with components

$$(\nabla f)_i = \frac{\partial f}{\partial x_i} = f_{,i}\,, \tag{1.2.8}$$

where the last entry of the continued equality introduces the convention whereby a subscript comma is used to denote partial differentiation. The divergence of a vector field $\mathbf{v}(p)$, denoted by $\nabla\cdot\mathbf{v}$, is the scalar

$$\nabla\cdot\mathbf{v} = \frac{\partial v_i}{\partial x_i} = v_{i,i}\,. \tag{1.2.9}$$

The curl of a vector field $\mathbf{v}(p)$, denoted by $\nabla\times\mathbf{v}$, is a vector with components

$$(\nabla\times\mathbf{v})_i = e_{ijk}v_{k,j}\,. \tag{1.2.10}$$

The Laplacian operator acting on a scalar field $f$, defined by $\nabla \cdot \nabla f$ and denoted by $\nabla^2 f$, is

$$\nabla^2 f = f_{,ii} \tag{1.2.11}$$

in indicial notation. If the gradient operator

$$\nabla = \mathbf{e}_i \frac{\partial}{\partial x_i} \tag{1.2.12}$$

is viewed as a vector, then useful generalizations of the foregoing results become obvious. For example, the gradient of a vector from the left or the right is the dyad $\nabla \mathbf{v}$ or $\mathbf{v}\nabla$ which is a second-order tensor with rectangular components

$$(\nabla \mathbf{v})_{ij} = v_{j,i} \qquad \text{or} \qquad (\mathbf{v}\nabla)_{ij} = v_{i,j}\,. \tag{1.2.13}$$

Likewise, the divergence of a second-order tensor from the left $\nabla \cdot \mathbf{T}$ or from the right $\mathbf{T} \cdot \nabla$ is a vector with rectangular components

$$(\nabla \cdot \mathbf{T})_j = T_{ij,i} \qquad \text{or} \qquad (\mathbf{T} \cdot \nabla)_j = T_{ji,i}\,, \tag{1.2.14}$$

and so on.

Gauss's divergence theorem on integrals of vector fields plays a key role in continuum mechanics. To state the theorem in its simplest form, consider a vector field $\mathbf{v} = v_i \mathbf{e}_i$ over a region $R$ of the point space. Let $S$ be the closed boundary surface of $R$ and let $n_i$ be the outward unit normal to $S$. Then the divergence theorem states that

$$\int_R \nabla \cdot \mathbf{v}\, dR = \int_S \mathbf{n} \cdot \mathbf{v}\, dS \quad \text{or} \quad \int_R v_{i,i}\, dR = \int_S n_j v_j\, dS\,. \tag{1.2.15}$$

Again, the result is readily generalized to tensor fields of second- and higher order if the divergence and inner product operators are properly interpreted. For the integral of the divergence from the left of the second-rank tensor field $\mathbf{T}$, for example, the divergence theorem states that

$$\int_R \nabla \cdot \mathbf{T}\, dR = \int_S \mathbf{n} \cdot \mathbf{T}\, dS \quad \text{or} \quad \int_R T_{ij,i}\, dR = \int_S T_{ij} n_i\, dS\,. \tag{1.2.16}$$

Most of the fields to be introduced to represent physical quantities depend on time $t$ as well as on position in a three-dimensional space. In a few situations, it is advantageous to treat the time coordinate in the same way as a spatial coordinate in a four-dimensional point space with coordinates $(p, t)$, as in Section 5.2. For the most part, however, the time coordinate is treated separately from spatial coordinates in applying the above conventions. Partial differentiation of a time dependent scalar field $f(p, t)$ is denoted by

$$\frac{\partial f}{\partial t} = f_{,t} \,. \tag{1.2.17}$$

### 1.2.2 Balance equations

Consider a deformable body occupying the region $R$ of the three-dimensional point space at time $t$. The closed boundary surface of $R$ is $S$ with outward unit normal vector $n_i$. The components of the second-order Cauchy stress tensor field with respect to the rectangular reference frame are denoted by $\sigma_{ij}(p, t)$. Consider any surface in $R$, possibly the boundary surface, and let $\nu_i$ be a unit normal vector at a particular point on the surface. According to the Cauchy stress principle, the vector representing local traction, or force per unit area, as a function of position on the surface is

$$\mathcal{T}_i = \sigma_{ji}\nu_j \,. \tag{1.2.18}$$

This is the traction exerted on the material on the side of the surface in the direction of $-\nu_i$. If $S$ is the boundary surface, this traction may arise from external sources. For a surface interior to $R$, the relationship describes the action of the material on one side of the surface on the material on the other side. The validity of this relationship between the stress tensor and the traction vector does not require that the fields be equilibrium fields. No couple stress effects are included in the discussion of fracture mechanics in this book.

Suppose that a particular fixed configuration of the body is identified as a reference configuration. Material particles are labeled by associating each with its position in the reference configuration, say the point with coordinates $x_1^o, x_2^o, x_3^o$, represented collectively by $p^o$. The deformation is specified by a nonsingular mapping of points $x_i^o \mathbf{e}_i$

in the reference configuration to points $x_i\mathbf{e}_i$ at time $t$ in the "current" configuration

$$x_i = \chi_i(x_1^o, x_2^o, x_3^o, t) \tag{1.2.19}$$

with inverse

$$x_i^o = \chi_i^o(x_1, x_2, x_3, t)\,. \tag{1.2.20}$$

The particle displacement $\mathbf{u} = u_i\mathbf{e}_i$ is given by

$$u_i = \chi_i(x_1^o, x_2^o, x_3^o, t) - x_i^o \quad \text{or} \quad u_i = x_i - \chi_i^o(x_1, x_2, x_3, t) \tag{1.2.21}$$

in a material description or a spatial description , respectively, of the deformation. The particle velocity field $\mathbf{v} = v_i\mathbf{e}_i$ is given by

$$v_i = \frac{\partial \chi_i}{\partial t} = \left.\frac{\partial u_i}{\partial t}\right|_{p^o}, \tag{1.2.22}$$

which is naturally a function of $(p^o, t)$ but which can be viewed as a function of $(p, t)$ through (1.2.20). The particle acceleration $\mathbf{a} = a_i\mathbf{e}_i$ is given by either

$$a_i = \left.\frac{\partial v_i}{\partial t}\right|_{p^o} = \left.\frac{\partial^2 u_i}{\partial t^2}\right|_{p^o} \tag{1.2.23}$$

or

$$a_i = \left.\frac{\partial v_i}{\partial t}\right|_{p} + v_j\frac{\partial v_i}{\partial x_j}\,. \tag{1.2.24}$$

The two expressions are both forms of the material time derivative of particle velocity, the first in the material description of motion and the second in the spatial description. The material time derivative will be denoted by a superposed dot, that is, $\mathbf{a} = \dot{\mathbf{v}}$, which implies that $a_i = \dot{v}_i$ in the rectangular frame.

Suppose that the mass density of the material occupying $R$ is $\rho(p, t)$. Conservation of mass during a deformation with particle velocity field $v_i$ requires that

$$\dot{\rho} + \rho v_{i,i} = 0\,. \tag{1.2.25}$$

If the material is subjected to a body force with rectangular components $f_i(p, t)$ per unit mass, the balance of linear momentum for an arbitrary *material* volume requires that the fields must satisfy

$$\frac{\partial \sigma_{ji}}{\partial x_j} + \rho f_i = \rho a_i\,. \tag{1.2.26}$$

The balance of angular momentum requires that the stress tensor is symmetric, or

$$\sigma_{ij} = \sigma_{ji} \,. \tag{1.2.27}$$

If traction and particle velocity are discontinuous across a surface moving through the material, then the momentum balance equation (1.2.26) is replaced by

$$[\![\sigma_{ij}\nu_i + \rho V v_j]\!] = 0 \,, \tag{1.2.28}$$

where $V$ is the local speed of the surface in the direction of the unit normal $\nu_i$ with respect to material particles instantaneously on the surface. The double square bracket notation is adopted for the jump in the value of a function $f$ that is discontinuous across a surface whereby

$$[\![f]\!] = f^+ - f^- \,, \tag{1.2.29}$$

where $f^\pm$ is the limiting value of $f$ on the surface as it is approached in the direction of $\mp\nu_i$.

The instantaneous rate of work of a traction distribution on a surface is defined in the following way. Suppose that $S$ is a *material* surface in the body, that is, material particles on the surface at any one time are on the surface at all times. If $\mathcal{T}_i$ is the traction distribution on the surface and $v_i$ is the velocity of material particles on the surface then the instantaneous rate of work being done on the material on the $-\nu_i$ side of $S$ by the traction is

$$\int_S \mathcal{T}_i v_i \, dS = \int_S \sigma_{ji}\nu_j v_i \, dS \,. \tag{1.2.30}$$

The surface $S$ may be an interior surface or any part of the boundary surface of the body.

The kinetic energy per unit volume is defined as

$$T = \tfrac{1}{2}\rho \mathbf{v}\cdot\mathbf{v} = \tfrac{1}{2}\rho v_i v_i \,, \tag{1.2.31}$$

where $\rho$ is the material mass density. Total kinetic energy of a material volume, say $T_{tot}$, is the integral of $T$ over the volume. The stress power per unit volume is defined as

$$P = \sigma_{ij}\frac{\partial v_i}{\partial x_j} \,. \tag{1.2.32}$$

Suppose that a certain volume of material instantaneously occupies a region of space $R$, and that the boundary surface of the region is $S$ with outward normal $n_i$. Then the rate of work being done on the body by the traction on $S$ and by the body force $f_i$ throughout $R$ is equal to the net stress power plus the rate of increase of kinetic energy throughout $R$, or

$$\int_S \sigma_{ij} n_i v_j \, dS + \int_R \rho f_j v_j \, dR = \int_R (P + \dot{T}) \, dR . \tag{1.2.33}$$

This relation can be established by applying the divergence theorem (1.2.16) to the first term on the left side, and by incorporating the momentum balance equations (1.2.26), conservation of mass (1.2.25), and the kinematic definitions.

The field equations and integral relations introduced above with respect to the current configuration of the deformable body can be transformed into equivalent expressions with respect to the reference configuration. Suppose that the body occupies the region $R_o$ of space with boundary surface $S_o$ and outward normal $n_i^o$ in its reference configuration. The mass density in the reference configuration is $\rho_o$. The nonsymmetric nominal stress tensor field (or the transpose of the first Piola-Kirchhoff stress tensor) is related to the Cauchy stress by

$$\sigma_{ij}^o = \frac{\rho_o}{\rho} \frac{\partial \chi_i^o}{\partial x_k} \sigma_{kj} . \tag{1.2.34}$$

If $f_i^o$ is the body force per unit mass in the reference configuration, then the balance of linear momentum takes the form

$$\frac{\partial \sigma_{ij}^o}{\partial x_i^o} + \rho_o f_j^o = \rho_o \frac{\partial^2 \chi_j}{\partial t^2} . \tag{1.2.35}$$

Similarly, the work rate balance equation takes the form

$$\int_{S_o} \sigma_{ij}^o n_i^o \frac{\partial \chi_j}{\partial t} \, dS_o + \int_{R_o} \rho_o f_j^o \frac{\partial \chi_j^o}{\partial t} \, dR_o =$$

$$\int_{R_o} \sigma_{ij}^o \frac{\partial^2 \chi_j}{\partial x_i^o \partial t} \, dR_o + \frac{\partial}{\partial t} \int_{R_o} \tfrac{1}{2} \rho_o \frac{\partial \chi_i}{\partial t} \frac{\partial \chi_i}{\partial t} \, dR_o , \tag{1.2.36}$$

where $T_o = \frac{1}{2}\rho_o \chi_{i,t}\, \chi_{i,t}$ is the kinetic energy per unit volume in the reference configuration. Because the reference configuration is fixed, differentiation with respect to time commutes with integration over $R_o$. Thus, the first term on the right side of (1.2.36) can be rewritten as

$$\int_{R_o} \sigma^o_{ij} \frac{\partial^2 \chi_j}{\partial x_i \partial t}\, dR_o = \frac{\partial}{\partial t} \int_{R_o} U_o\, dR_o\,, \tag{1.2.37}$$

where $U_o$ is the accumulated stress work per unit volume, or the stress work density

$$U_o = \int_{-\infty}^{t} \sigma^o_{ij} \frac{\partial^2 \chi_j}{\partial x^o_i \partial \tau}\, d\tau \tag{1.2.38}$$

at each material point in the reference configuration.

Suppose that $\omega(p,t)$ is a time-dependent field which represents the density per unit mass of some physical quantity throughout a deformable body. Then the net measure of the quantity over a volume of material occupying a time-dependent region of space $R(t)$ is

$$\Omega(t) = \int_{R(t)} \rho(p,t)\omega(p,t)\, dR\,. \tag{1.2.39}$$

Suppose that $S(t)$ is the time dependent boundary surface of $R(t)$ with outward normal $n_i$, and that the velocity of any point on $S$ is $v^S_i$. The time rate of change of the net measure of $\Omega$ for the material instantaneously in $R(t)$ is

$$\dot{\Omega} = \int_{R(t)} \frac{\partial(\rho\omega)}{\partial t}\, dR + \int_{S(t)} \rho\omega V_n\, dS\,, \tag{1.2.40}$$

where $V_n = v^S_i n_i$ is the local normal velocity of the surface $S$ (Truesdell and Toupin 1960). The result states that the value of $\Omega$ for the material in the region $R(t)$ changes for two reasons. The first term on the right side of (1.2.40) is a contribution due to the pointwise rate of change of density $\omega$ throughout $R$. The second term on the right side of (1.2.40) accounts for the fact that, as the surface $S$ moves, there is a local flux of $\omega$ through the boundary. The net value of this flux also contributes to the rate of change of $\Omega$. In the special case when $V_n$ is the component of velocity of the material particles on $S(t)$ in the direction of $n_i$, or $V_n = v_i n_i$, the surface $S$ is a material surface and the

result (1.2.40) is an expression for the rate of change of $\Omega$ for a fixed material volume. In the latter case, (1.2.40) is called the Reynold's transport theorem. For the general case when the normal speed $V_n$ has no connection to the motion of the material particles currently on the surface, the result will be called the *generalized transport theorem.* This theorem plays a central role in computing energy flux through a tubular surface surrounding a crack edge and moving with it, for example.

If the same volume of material is considered in the reference configuration of the body, then the statement of the generalized transport theorem has the form

$$\dot{\Omega}(t) = \int_{R_o} \rho_o \dot{\omega}_o \, dR_o + \int_{S_o} \rho_o \omega_o V_n^o \, dS_o \,, \tag{1.2.41}$$

where $V_n^o = (v_i^S - v_i) n_k^o \chi_{k,i}^o$ is the normal speed of the surface $S$ mapped back to the reference configuration. The normal speed was introduced as an absolute speed of the surface, so in the reference configuration it assumes the form of a speed with respect to the speed of the particles currently on $S_o$.

Finally, some specific deformation measures are adopted. The rate of deformation tensor $\mathbf{d}$ is the symmetric part of the particle velocity gradient in the current configuration, and its rectangular components are

$$d_{ij} = \frac{1}{2}\left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i}\right) . \tag{1.2.42}$$

The strain tensor $\mathbf{E}$ in terms of displacement (1.2.21) as a function of coordinates in the reference configuration has components

$$E_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j^o} + \frac{\partial u_j}{\partial x_i^o} + \frac{\partial u_k}{\partial x_i^o}\frac{\partial u_k}{\partial x_j^o}\right) . \tag{1.2.43}$$

Likewise, the strain tensor $\mathbf{E}^*$ in terms of displacement components as a function of coordinates in the current configuration has components

$$E_{ij}^* = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} + \frac{\partial u_k}{\partial x_i}\frac{\partial u_k}{\partial x_j}\right) . \tag{1.2.44}$$

When all components of displacement gradients in either description are small, the terms involving products of gradient components in

(1.2.43) and (1.2.44) will be small compared to the terms linear in gradient components. The two strain expressions then have the same form, except that the gradients are taken with respect to material coordinates in the former case and spatial coordinates in the latter case. When the displacements and the displacement gradients are sufficiently small, the distinction between the two descriptions is ignored and a single set of coordinates $x_1, x_2, x_3$ is used. In this case, the strain tensor components are

$$\epsilon_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) . \tag{1.2.45}$$

A description of deformation based on (1.2.45) is referred to as a small strain formulation or an infinitesimal strain formulation. Although the small strain formulation can be viewed as an approximation to a theory of deformable bodies undergoing arbitrary deformations, the process of extracting the small strain formulation from the general theory of deformation as an asymptotic result is somewhat ambiguous (Gurtin 1972; Malvern 1969). An interesting perspective on the conditions necessary for the small strain conditions to be met for elastic-plastic materials is given by Rice et al. (1979). Formulations based on small strain concepts are themselves mathematically complete, however, and they provide a consistent framework for solving boundary value problems in mechanics. Most of the results in this book are based on small strain concepts. The few exceptions are clearly identified.

### *1.2.3 Linear elastodynamics*

The fundamental set of field equations governing the motion of a homogeneous and isotropic elastic body consists of the strain displacement relation for small strain (1.2.45)

$$\boldsymbol{\epsilon} = \tfrac{1}{2}\left(\nabla\mathbf{u} + \mathbf{u}\nabla\right) \quad \text{or} \quad \epsilon_{ij} = \tfrac{1}{2}\left(u_{j,i} + u_{i,j}\right) , \tag{1.2.46}$$

the momentum balance equations (1.2.26) and (1.2.27)

$$\nabla \cdot \boldsymbol{\sigma} + \rho\mathbf{f} = \rho\ddot{\mathbf{u}} \quad \text{or} \quad \sigma_{ij,i} + \rho f_j = \rho\ddot{u}_j , \tag{1.2.47}$$

and the linear stress–strain relation

$$\boldsymbol{\sigma} = \lambda\mathbf{I}\,\mathrm{tr}(\boldsymbol{\epsilon}) + 2\mu\boldsymbol{\epsilon} \quad \text{or} \quad \sigma_{ij} = \lambda\delta_{ij}\epsilon_{kk} + 2\mu\epsilon_{ij} , \tag{1.2.48}$$

where $\lambda$ and $\mu$ are the positive Lamé elastic constants. The field equations are to be satisfied throughout a region of space occupied by the elastic solid in its initial, undeformed configuration. No distinction is made between the deformed and undeformed configurations in the linear theory. Thus, fields representing the same physical quantities in the current "deformed" configuration (without superscript or subscript "o") and in the reference "undeformed" configuration in Section 1.2.2 are viewed as being indistinguishable, except that a displacement vector is associated with each material particle. The superscript or subscript "o" designation is dropped in the linear formulation. Thus, for example, the rate of work balance (1.2.33) or (1.2.36) takes the form

$$\int_S \sigma_{ij} n_i \frac{\partial u_j}{\partial t}\, dS + \int_R \rho f_j \frac{\partial u_j}{\partial t}\, dR = \frac{\partial}{\partial t} \int_R (U + T)\, dR\,, \qquad (1.2.49)$$

where the body occupies region $R$ with boundary $S$, $T$ is the kinetic energy density, and $U$ is the stress work density for general material response. In the case of linear elasticity, $U$ is the elastic strain energy density $\frac{1}{2}\sigma_{ij}\epsilon_{ij}$.

If the momentum equation is written in terms of displacement $\mathbf{u}$ by substituting first from (1.2.48) and then from (1.2.46), Navier's equation of motion

$$(\lambda + \mu)\nabla(\nabla \cdot \mathbf{u}) + \mu \nabla^2 \mathbf{u} + \rho \mathbf{f} = \rho \ddot{\mathbf{u}} \qquad (1.2.50)$$

is obtained. In light of the vector identity $\nabla \times (\nabla \times \mathbf{u}) = \nabla(\nabla \cdot \mathbf{u}) - \nabla^2 \mathbf{u}$, (1.2.50) can be rewritten as

$$c_d^2 \nabla(\nabla \cdot \mathbf{u}) - c_s^2 \nabla \times (\nabla \times \mathbf{u}) + \mathbf{f} = \ddot{\mathbf{u}}\,, \qquad (1.2.51)$$

where

$$c_d = \sqrt{\frac{\lambda + 2\mu}{\rho}} \qquad \text{and} \qquad c_s = \sqrt{\frac{\mu}{\rho}}\,. \qquad (1.2.52)$$

Let the body force $\mathbf{f}$ be zero for the time being. The result of operating on each term of (1.2.51) with the divergence operator $\nabla\cdot$ is

$$c_d^2 \nabla^2 (\nabla \cdot \mathbf{u}) = (\nabla \cdot \mathbf{u})_{,tt}\,. \qquad (1.2.53)$$

Therefore, the dilatation $\nabla \cdot \mathbf{u}$ satisfies the elementary wave equation with characteristic wave speed $c_d$, called the dilatational wave speed. Similarly, the result of operating on each term of (1.2.51) with the curl operator $\nabla\times$ is

$$c_s^2 \nabla^2 (\nabla \times \mathbf{u}) = (\nabla \times \mathbf{u})_{,tt} \tag{1.2.54}$$

so that the rotation vector $\frac{1}{2}\nabla \times \mathbf{u}$ satisfies the elementary wave equation with characteristic wave speed $c_s$, called the shear wave speed or the rotational wave speed. In an unbounded isotropic elastic solid, waves of pure dilatation (rotation) propagate at normal speed $c_d$ ($c_s$). Furthermore, the two types of wave are independent in the interior of a solid. At boundaries, such as the faces of a crack, for example, the two types of wave interact. This is the feature of elastodynamic fields that accounts for most of the complication in solving initial-boundary value problems. A complete discussion of plane wave propagation in unbounded bodies and plane wave reflection from boundaries is given by Achenbach (1973). He also presents the basic singular solutions of elastodynamics, principally, Stokes's solution for a time-dependent concentrated force in an unbounded solid and Love's solution for a center of dilatation of time-dependent strength in an unbounded solid.

The dilatational and shear wave speeds for an isotropic elastic material are defined in (1.2.52) in terms of mass density and the Lamé elastic constants $\lambda$ and $\mu$. It is sometimes more convenient to express the wave speeds in terms of Young's modulus $E$ and Poisson's ratio $\nu$. The relationships among these elastic constants are

$$E = \frac{\mu(3\lambda + 2\mu)}{\lambda + \mu}, \qquad \nu = \frac{\lambda}{2(\lambda + \mu)} \tag{1.2.55}$$

which can be inverted to yield

$$\lambda = \frac{E\nu}{(1+\nu)(1-2\nu)}, \qquad \mu = \frac{E}{2(1+\nu)}. \tag{1.2.56}$$

In terms of $E$ and $\nu$, the wave speeds are

$$c_d = \sqrt{\frac{E(1-\nu)}{\rho(1+\nu)(1-2\nu)}}, \qquad c_s = \sqrt{\frac{E}{2\rho(1+\nu)}}. \tag{1.2.57}$$

Another useful relationship follows from recognizing that the ratio $c_s/c_d$ has a form depending only on Poisson's ratio,

$$\frac{c_s^2}{c_d^2} = \frac{1-2\nu}{2(1-\nu)}. \tag{1.2.58}$$

To formulate an initial-boundary value problem in elastodynamics, an appropriate set of initial and boundary conditions is required. In view of the Navier equation, values of $\mathbf{u}$ and $\dot{\mathbf{u}}$ must be specified throughout the body initially, or some conditions equivalent to these must be specified. Boundary conditions must be specified on the boundary of the body for all time beyond the initial time. Conditions on the boundary surface may be:

i. the traction is specified,
ii. the displacement is specified,
iii. the component of traction in a certain direction is specified and the displacement in the plane perpendicular to this direction is also specified, or
iv. the component of displacement in a certain direction is specified and the traction in the plane perpendicular to this direction is also specified.

The boundary may be divided into any number of parts, but one and only one of the conditions (i) through (iv) must be specified over each part. For an appropriate set of initial and boundary conditions, Neumann's uniqueness theorem ensures that the initial-boundary value problem has at most one solution, provided that energy is added to the body or taken from it only through the loaded part of the boundary (Gurtin 1972; Achenbach 1973). This property of uniqueness of solutions has enormous practical value, for it means that an elastodynamic field obtained by any means whatsoever which satisfies the governing equations, initial conditions, and boundary conditions is the only solution of the problem. In obtaining solutions to elastic crack problems, conditions are often assumed in constructing a candidate solution that cannot be verified a priori. Nonetheless, if such a candidate solution can be shown to satisfy the governing equations and other conditions, then it is certain that it represents the unique solution. For dynamically growing cracks, the statement of the uniqueness theorem requires some modification due to the possibility of energy flow out of the body through the crack tip, as discussed in Section 7.7.

The elastodynamic reciprocal theorem provides a useful relationship between solutions of two different boundary value problems

for the same elastic body. Suppose that solutions of two different boundary value problems for a given body exist, say $\mathbf{u}$ and $\mathbf{u}^\star$. Furthermore, suppose that the displacement and particle velocity are zero for both solutions at the initial time, say $t = 0$. In addition, suppose that no body forces are present in either boundary value problem. Then, if $\mathbf{T}$ and $\mathbf{T}^\star$ are the tractions on the boundary surface, say $S$, corresponding to the two solutions, then the solutions are related by

$$\int_S \int_0^t \mathbf{T}(p_S, t-\tau) \cdot \mathbf{u}^\star(p_S, \tau)\, d\tau\, dS$$
$$= \int_S \int_0^t \mathbf{T}^\star(p_S, t-\tau) \cdot \mathbf{u}(p_S, \tau)\, d\tau\, dS\,. \qquad (1.2.59)$$

The generalization of Betti's reciprocal theorem to elastodynamics, including body forces and general initial conditions, is discussed in detail by Gurtin (1972) and Achenbach (1973). The theorem was extended to the case of unbounded domains by Wheeler and Sternberg (1968). There is no difficulty in extending the theorem to apply for general initial conditions and body forces.

The practical matter of integrating (1.2.51) to determine particular solutions often hinges on a representation of $\mathbf{u}$ in terms of other fields which satisfy the elementary wave equations with characteristic speed $c_d$ or $c_s$. The advantage gained through such a representation is that various means are available for solving the wave equation. A difficulty with this approach concerns the completeness of the representation. Does every solution of the Navier equation have a representation of the assumed form? This matter is discussed by Sternberg (1960) for the representations to be introduced here.

A complete representation of a solution of the wave equation is obtained by means of the Helmholtz additive decomposition of the displacement vector into the gradient of a scalar field $\phi$ and the curl of a divergence free vector field $\boldsymbol{\psi}$,

$$\mathbf{u} = \nabla\phi + \nabla \times \boldsymbol{\psi} \quad \text{or} \quad u_i = \phi_{,i} + e_{ijk}\psi_{k,j}\,, \qquad (1.2.60)$$

where the dilatational displacement potential $\phi$ and the shear displacement potential $\boldsymbol{\psi}$ are wave functions satisfying

$$c_d^2\nabla^2\phi - \ddot{\phi} = 0\,, \qquad c_s^2\nabla^2\boldsymbol{\psi} - \ddot{\boldsymbol{\psi}} = 0\,, \quad \nabla\cdot\boldsymbol{\psi} = 0\,. \qquad (1.2.61)$$

In terms of the displacement potential functions $\phi$ and $\boldsymbol{\psi}$, the dilatation of the deformation is $\nabla \cdot \mathbf{u} = \nabla^2 \phi$ and the rotation vector of the deformation is $\frac{1}{2}\nabla \times \mathbf{u} = -\frac{1}{2}\nabla^2 \boldsymbol{\psi}$. The relationship (1.2.60) is called the Lamé representation of the displacement vector.

Sternberg (1960) also discussed other representations. For example, the Poisson representation is

$$\mathbf{u} = \mathbf{u}^d + \mathbf{u}^s \,, \tag{1.2.62}$$

where

$$\nabla \times \mathbf{u}^d = 0 \,, \qquad \nabla \cdot \mathbf{u}^s = 0 \tag{1.2.63}$$

and

$$c_d^2 \nabla^2 \mathbf{u}^d - \ddot{\mathbf{u}}^d = 0 \,, \qquad c_s^2 \nabla^2 \mathbf{u}^s - \ddot{\mathbf{u}}^s = 0 \,. \tag{1.2.64}$$

This representation has been applied effectively in two-dimensional cases when the elastodynamic fields are steady with respect to a moving observer or are self-similar, so that the wave equations reduce to Laplace's equation for which a general solution is available. For the special case of antiplane shear deformation, $u_1 = u_2 = 0$ and $u_3 = w(x_1, x_2, t)$. The out-of-plane displacement satisfies the elementary wave equation

$$c_s^2 \nabla^2 w - \ddot{w} = 0 \,. \tag{1.2.65}$$

Yet another representation that is shown by Sternberg (1960) to be equivalent to the Lamé representation is

$$\mathbf{u} = 2(1-\nu)\left[c_d^2 \nabla^2 \mathbf{G} - \ddot{\mathbf{G}}\right] - \nabla(\nabla \cdot \mathbf{G}) \,, \tag{1.2.66}$$

where the vector fields $\mathbf{G}$ and $\mathbf{H}$ satisfy

$$c_s^2 \nabla^2 \mathbf{G} - \ddot{\mathbf{G}} = \mathbf{H} \,, \qquad c_d^2 \nabla^2 \mathbf{H} - \ddot{\mathbf{H}} = 0 \,. \tag{1.2.67}$$

The Lamé representation is used extensively in this book, and expressions for rectangular components of displacement and stress in terms of $\phi$ and components of $\boldsymbol{\psi}$ are recorded here for later reference. From (1.2.60), the components of displacement in terms of the

displacement potentials are

$$
\begin{aligned}
u_1 &= \frac{\partial \phi}{\partial x_1} + \frac{\partial \psi_3}{\partial x_2} - \frac{\partial \psi_2}{\partial x_3}\,, \\
u_2 &= \frac{\partial \phi}{\partial x_2} + \frac{\partial \psi_1}{\partial x_3} - \frac{\partial \psi_3}{\partial x_1}\,, \\
u_3 &= \frac{\partial \phi}{\partial x_3} + \frac{\partial \psi_2}{\partial x_1} - \frac{\partial \psi_1}{\partial x_2}\,.
\end{aligned}
\tag{1.2.68}
$$

If these expressions are substituted into (1.2.48), then the components of stress in terms of the displacement potentials are

$$
\begin{aligned}
\sigma_{11} &= \mu\left[\frac{c_d^2}{c_s^2}\nabla^2\phi - 2\frac{\partial^2\phi}{\partial x_2^2} - 2\frac{\partial^2\phi}{\partial x_3^2} + 2\frac{\partial^2\psi_3}{\partial x_1\partial x_2} - 2\frac{\partial^2\psi_2}{\partial x_3\partial x_1}\right]\,, \\
\sigma_{22} &= \mu\left[\frac{c_d^2}{c_s^2}\nabla^2\phi - 2\frac{\partial^2\phi}{\partial x_3^2} - 2\frac{\partial^2\phi}{\partial x_1^2} + 2\frac{\partial^2\psi_1}{\partial x_2\partial x_3} - 2\frac{\partial^2\psi_3}{\partial x_1\partial x_2}\right]\,, \\
\sigma_{33} &= \mu\left[\frac{c_d^2}{c_s^2}\nabla^2\phi - 2\frac{\partial^2\phi}{\partial x_1^2} - 2\frac{\partial^2\phi}{\partial x_2^2} + 2\frac{\partial^2\psi_2}{\partial x_3\partial x_1} - 2\frac{\partial^2\psi_1}{\partial x_2\partial x_3}\right]\,, \\
\sigma_{12} &= \mu\left[2\frac{\partial^2\phi}{\partial x_1\partial x_2} + \frac{\partial^2\psi_3}{\partial x_2^2} - \frac{\partial^2\psi_2}{\partial x_2\partial x_3} - \frac{\partial^2\psi_3}{\partial x_1^2} + \frac{\partial^2\psi_1}{\partial x_3\partial x_1}\right]\,, \\
\sigma_{23} &= \mu\left[2\frac{\partial^2\phi}{\partial x_2\partial x_3} + \frac{\partial^2\psi_1}{\partial x_3^2} - \frac{\partial^2\psi_3}{\partial x_3\partial x_1} - \frac{\partial^2\psi_1}{\partial x_2^2} + \frac{\partial^2\psi_2}{\partial x_1\partial x_2}\right]\,, \\
\sigma_{31} &= \mu\left[2\frac{\partial^2\phi}{\partial x_3\partial x_1} + \frac{\partial^2\psi_2}{\partial x_1^2} - \frac{\partial^2\psi_1}{\partial x_1\partial x_2} - \frac{\partial^2\psi_2}{\partial x_3^2} + \frac{\partial^2\psi_3}{\partial x_2\partial x_3}\right]\,.
\end{aligned}
\tag{1.2.69}
$$

Although no specific solutions are presented for anisotropic elastic material response, some of the general developments are carried out for an elastic material of general symmetry. The most general stress–strain relation for a material possessing a strain energy function is

$$
\sigma_{ij} = C_{ijkl}\epsilon_{kl}\,. \tag{1.2.70}
$$

The array of elastic constants satisfies the symmetry conditions

$$C_{ijkl} = C_{klij} = C_{jikl} = C_{ijlk} \tag{1.2.71}$$

so that there are 21 elastic constants at most. In terms of the stiffness array $C_{ijkl}$, the strain energy density of the material is

$$U = \tfrac{1}{2} C_{ijkl} \epsilon_{ij} \epsilon_{kl} = \tfrac{1}{2} C_{ijkl} u_{i,j}\, u_{k,l} \ . \tag{1.2.72}$$

It is assumed that the relationship is invertible, so that there exists an array $M_{ijkl}$ such that

$$\epsilon_{ij} = M_{ijkl} \sigma_{kl} \ . \tag{1.2.73}$$

The array $M_{ijkl}$ of elastic compliances possesses the same symmetries as $C_{ijkl}$ does.

### 1.2.4 Inelastic materials

The limited results available on dynamic fracture of inelastic materials are also discussed in the chapters to follow. General analytical procedures do not exist for solving boundary value problems involving inelastic material response, except for the case of linear viscoelasticity, and results are obtained on the basis of ad hoc procedures. The relevant field equations for formulating such problems, aside from the constitutive equations, are included in Section 1.2.2. Field equations describing material response are presented along with the specific problems studied. Among these are the determination of asymptotic crack tip fields for crack growth in a rate independent elastic-plastic material in Sections 4.4 and 4.5. Crack tip fields for rapid crack growth in an elastic-viscous material are considered in Section 4.6, and in an elastic-viscoplastic material in Section 4.7.

Chapter 8 is devoted to the study of plasticity and strain-rate effects in dynamic fracture. Here, a few results of fairly broad applicability are obtained for rapid crack growth in elastic-plastic and elastic-viscoplastic materials. This chapter also includes brief discussions of ductile void growth in plastic materials and the apparent rate sensitivity exhibited by brittle materials undergoing rapid microcracking. In all cases, constitutive models are developed as part of the problem formulation.

## 1.3 Analytic functions and Laplace transforms

Some basic mathematical results to be used subsequently are summarized in this section. These results concern the general properties of an analytic function of a complex variable and the properties of Laplace transforms, both the one-sided and the two-sided types. Theorems are stated without proof, and only to a degree of generality important to the applications to follow.

The theory of analytic functions of a complex variable is fully developed by Hille (1959) and, within the context of applications, by Carrier, Krook, and Pearson (1966). A useful summary of main results is given by Seebass (1983). Integral transforms are developed by Carrier et al. (1966), and aspects relevant to the Wiener-Hopf method are summarized by Noble (1958).

### *1.3.1 Analytic functions of a complex variable*

Consider the function $f(\zeta)$ of the complex variable $\zeta = x + iy = re^{i\theta}$ defined in a neighborhood of a particular point. If $f(\zeta)$ is differentiable with respect to $\zeta$ at that point, then it is an analytic function at that point. If $f(\zeta)$ is analytic at every point in a region, it is said to be analytic in the region. Analyticity of $f(\zeta)$, in turn, implies that derivatives of all orders exist.

If $f(\zeta)$ is analytic then, with $f(\zeta) = u(x,y) + iv(x,y)$, it follows that

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}. \tag{1.3.1}$$

These relations are called the Cauchy-Riemann equations. If $f(\zeta)$ is analytic in a region, then the Cauchy-Riemann relations imply that

$$\nabla^2 u = 0, \qquad \nabla^2 v = 0 \tag{1.3.2}$$

in that region. Real functions that satisfy Laplace's equation are said to be harmonic functions. A general solution of the partial differential equation $\nabla^2 u = 0$ over some region is

$$u = \mathrm{Re}\{F(\zeta)\} \quad \text{or} \quad u = \mathrm{Im}\{G(\zeta)\}, \tag{1.3.3}$$

where $F(\zeta)$ or $G(\zeta)$ is an analytic function in the region that must be determined from the boundary conditions.

Among the most useful results on analytic functions for purposes of applied analysis are Cauchy's integral theorem and its consequences. If $f(\zeta)$ is analytic on and inside a simple closed curve $C$ in the complex $\zeta$-plane, then

$$\oint_C f(\zeta)\, d\zeta = 0\,. \tag{1.3.4}$$

Similarly, if $f(\zeta)$ is analytic on and outside of a simple closed curve, including at infinity, and if $\left|f(re^{i\theta})\right| = o(1/r)$ as $r \to \infty$ then (1.3.4) also applies. Either form of the theorem can be extended to apply for multiple simple closed curves by interconnecting the simple curves with paths through regions of analyticity. If these paths are traversed in both directions, then the multiple curves and interconnections can be viewed as a simple closed curve.

A consequence of Cauchy's theorem is Cauchy's integral formula. Suppose that $f(\zeta)$ is analytic on and inside a simple curve $C$ enclosing a region of the $\zeta$-plane. If $\zeta_o$ is a point inside $C$ then

$$f(\zeta_o) = \frac{1}{2\pi i} \oint_C \frac{f(\zeta)}{\zeta - \zeta_o}\, d\zeta\,, \tag{1.3.5}$$

where the path is traversed in the counterclockwise direction. Similarly, suppose that $f(\zeta)$ is analytic on and outside of a simple closed curve $C$ in the $\zeta$-plane, including at infinity. Then, if $\left|f(re^{i\theta})\right| = o(1)$ as $r \to \infty$ and if $\zeta_o$ is outside of $C$ then

$$f(\zeta_o) = -\frac{1}{2\pi i} \oint_C \frac{f(\zeta)}{\zeta - \zeta_o}\, d\zeta\,, \tag{1.3.6}$$

where the curve $C$ is counterclockwise oriented.

Various other extensions of the integral formula (1.3.5) are useful in different circumstances. For example, if $f(\zeta)$ is analytic on and inside the simple closed curve $C$ and $\zeta_o$ is inside $C$ then

$$f^{(n)}(\zeta_o) = \frac{n!}{2\pi i} \oint_C \frac{f(\zeta)}{(\zeta - \zeta_o)^{n+1}}\, d\zeta\,,$$

where the left side denotes the $n$-th derivative of $f(\zeta)$. Cauchy's integral formula can also be used to establish other useful properties of

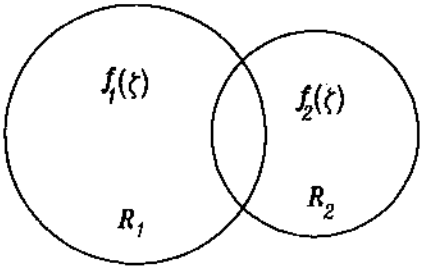


Figure 1.3. Overlapping regions in the complex $\zeta$-plane used to illustrate the role of the identity theorem in analytic continuation.

analytic functions. For example, according to the maximum modulus theorem, if $f(\zeta)$ is analytic on and inside a simple closed curve $C$, then either the maximum value of $|f(\zeta)|$ is attained only on $C$ or $f(\zeta)$ is constant throughout the region. Another very powerful result is Liouville's theorem, which states that a bounded entire function of a complex variable is necessarily a constant. A generalization of Liouville's theorem is stated in Section 2.5.2 in connection with the Wiener-Hopf technique. Yet another result, called the zero counting formula, is used to great advantage in Section 2.5.

A theorem which provides the basis for techniques of *analytic continuation* is the identity theorem or uniqueness theorem of analytic function theory. Suppose that $f(\zeta)$ is analytic in a region of the complex $\zeta$-plane and that $f(\zeta) = 0$ on any dense subset of that region, for example, along a segment of a curve or over a small area. Under these conditions, $f(\zeta)$ is identically zero throughout the region. This very important theorem plays a central role in most techniques of applied mathematics based on analytic function theory and on integral transforms. It follows immediately from the identity theorem that if $f_1(\zeta)$ and $f_2(\zeta)$ are both analytic throughout a region of the $\zeta$-plane and if $f_1(\zeta) = f_2(\zeta)$ on a dense subset of the region, then the two functions are equal to each other and, indeed, each represents precisely the same function over the whole region.

To see how the identity theorem provides the basis for analytic continuation, consider functions $f_1(\zeta)$ and $f_2(\zeta)$ that are analytic in regions $R_1$ and $R_2$, respectively, of the complex plane. Furthermore, suppose that the two regions have some overlap, as shown schematically in Figure 1.3. Then, if $f_1(\zeta) = f_2(\zeta)$ within the common region of overlap, $f_1(\zeta)$ or $f_2(\zeta)$ represents the analytic continuation of

$f_2(\zeta)$ or $f_1(\zeta)$ into $R_1$ or $R_2$, respectively, and the functions together represent a single analytic function throughout the combined region $R_1 + R_2$. This analytic continuation idea is the basis for an important step in the Wiener-Hopf technique.

### 1.3.2 Laplace transforms

Laplace transforms are used in a rather formal way to obtain solutions to boundary value problems in the chapters to follow. The transforms are defined here, and certain properties which are exploited later are identified. The equations to be solved are typically partial differential equations involving functions of time and one or more spatial variables. For this general discussion, it is sufficient to consider a function of time $t$ and a single spatial variable, say $x$.

Consider a function $w(x,t)$ defined for $t > 0$ and $-\infty < x < \infty$. The Laplace transform on time of $w$ is denoted by a superposed hat on the function and is defined by

$$\widehat{w}(x,s) = \int_0^\infty w(x,t)\, e^{-st}\, dt\,, \tag{1.3.7}$$

where $s$ is the transform parameter. Suppose that $w(x,t)$ is integrable over any finite interval of $t$ and $|w(x,t)| = O\left(e^{\sigma_+ t}\right)$ as $t \to \infty$, where $\sigma_+$ is a real constant. Then the integral in (1.3.7) converges for $\mathrm{Re}(s) > \sigma_+$ and it defines some function of $s$ in that half plane. Furthermore, the integral can be differentiated with respect to $s$, and the derivative converges for $\mathrm{Re}(s) > \sigma_+$. Thus, $\widehat{w}(x,s)$ is an analytic function of the complex variable $s$ in $\mathrm{Re}(s) > \sigma_+$ for fixed $x$. In view of the identity theorem for analytic functions, it is sufficient to view $\widehat{w}(x,s)$ as a function of a real variable $s$ over some segment of the real axis in the half plane of analyticity. Once $\widehat{w}(x,s)$ is determined as an explicit function of $s$ in the course of solving the transformed differential equations, the definition of $\widehat{w}(x,s)$ can be extended to the entire complex $s$-plane, except for isolated singular points, by analytic continuation.

The function $w(x,t)$ is obtained by means of the transform inversion integral

$$w(x,t) = \frac{1}{2\pi i} \int_{\sigma - i\infty}^{\sigma + i\infty} \widehat{w}(x,s)\, e^{st}\, ds\,, \tag{1.3.8}$$

where $\sigma$ is any real parameter greater than $\sigma_+$. The integration path is a line parallel to the imaginary axis in the $s$-plane and within the region of analyticity of $\widehat{w}(x,s)$.

In the particular case when a transform is factorable into a product of transforms, say

$$\widehat{w}(x,s) = \widehat{w}_1(x,s)\widehat{w}_2(x,s)\,, \tag{1.3.9}$$

the function $w(x,t)$ itself is a convolution of $w_1(x,t)$ and $w_2(x,t)$, that is,

$$w(x,t) = \int_0^t w_1(x,t-\tau)w_2(x,\tau)\,d\tau\,. \tag{1.3.10}$$

Transform inversion integrals of the form (1.3.8) are often evaluated by changing the path of integration from the infinite line parallel to the imaginary axis to a path in a left half plane that encloses all of the singular points of the integrand in the complex $s$-plane. The change in path is accomplished by forming a closed path made up of the inversion path and the path enclosing the singularities plus circular arcs of indefinitely large radius, and then invoking Cauchy's theorem (1.3.4). For this process to be successful, it is essential that the integrals taken along the circular arcs of large radius vanish as the radius becomes indefinitely large. This will be the case provided that $\widehat{w}(x,s) \to 0$ uniformly as $|s| \to \infty$. This result follows from Jordan's lemma as applied to integrals of the form (1.3.8) (Seebass 1983).

The transform of the time derivative $\partial w(x,t)/\partial t$ is

$$\int_{0^+}^{\infty} \frac{\partial w}{\partial t} e^{-st}\,dt = s\widehat{w}(x,s) - w(x,0^+)\,. \tag{1.3.11}$$

If $w$ is identically zero for $t<0$ and if it increases discontinuously at $t=0$ to $w(x,0^+)$ then, in the sense of distributions (Stakgold 1968),

$$\int_{0^-}^{\infty} \frac{\partial w}{\partial t} e^{-st}\,dt = s\widehat{w}(x,s)\,. \tag{1.3.12}$$

The utility of the Laplace transform methods in solving linear differential equations is represented by (1.3.11). Multiplication of each term in the differential equation by $e^{-st}$ and integration over $t$ from $0^+$ to $\infty$, in effect, replaces time differentiation with multiplication by the

parameter $s$. The parameter $s$ can be manipulated algebraically. Once a solution of the transformed equations is found, the solution in the time domain is obtained by superposition over the continuous range of $s$ represented by (1.3.8). The completeness of the representation of functions in terms of integral transforms is established by means of Fourier's integral theorem (Carrier et al. 1966). In the present case, completeness implies that every function (within an appropriate class of integrable functions) has the representation (1.3.8), where the density function $\widehat{w}(x,s)$ is given by (1.3.7).

Consider now the function $\widehat{w}(x,s)$ for any fixed real $s$ in the region of convergence of $\widehat{w}(x,s)$ and for $-\infty < x < \infty$. The two-sided or bilateral Laplace transform of $\widehat{w}(x,s)$ is denoted by a corresponding uppercase letter and it is defined by the integral

$$\begin{aligned} W(\xi,s) &= \int_{-\infty}^{\infty} \widehat{w}(x,s)\, e^{-\xi x}\, dx \\ &= \int_{-\infty}^{0} \widehat{w}(x,s)\, e^{-\xi x}\, dx + \int_{0}^{\infty} \widehat{w}(x,s)\, e^{-\xi x}\, dx\,, \end{aligned} \tag{1.3.13}$$

where $\xi$ is the complex transform parameter. Suppose that $\widehat{w}(x,s)$ is integrable over any finite interval of $x$ and that

$$|\widehat{w}(x,s)| = \begin{cases} O\left(e^{\tau_+ x}\right) & \text{as } x \to +\infty, \\ O\left(e^{-\tau_- x}\right) & \text{as } x \to -\infty, \end{cases} \tag{1.3.14}$$

where $\tau_+$ and $\tau_-$ are real constants. The integral over positive values of $x$ in (1.3.13) then converges and defines an analytic function of $\xi$ in the half plane $\mathrm{Re}(\xi) > \tau_+$, whereas the integral over negative values of $x$ converges and defines an analytic function in the half plane $\mathrm{Re}(\xi) < \tau_-$. Thus, $W(\xi,s)$ exists and represents an analytic function in the strip $\tau_+ < \mathrm{Re}(\xi) < \tau_-$. Once sufficient information is available about the dependence of $W(\xi,s)$ on $\xi$ from the boundary value problem, the definition can be extended to the entire complex plane, except for isolated singular points, by means of analytic continuation. If $\tau_+ = \tau_-$ the strip of analyticity degenerates to a line, and if $\tau_+ > \tau_-$ then the two-sided transform does not exist.

The inversion integral for the two-sided Laplace transform is

$$\widehat{w}(x,s) = \frac{1}{2\pi i} \int_{\tau - i\infty}^{\tau + i\infty} W(\xi,s)\, e^{\xi x}\, d\xi\,, \tag{1.3.15}$$

where $\tau$ is a real parameter in the interval $\tau_+ < \tau < \tau_-$. The integration path is a line parallel to the imaginary axis in the $\xi$-plane within the strip of analyticity of $W(\xi, s)$.

Consider the special case of a function of $x$ and $t$ that is zero for all $x$ and $t < 0$ and that vanishes for all $x \leq 0$ and $t > 0$, say $w_+(x,t)$. The Laplace transform on time $\widehat{w}_+(x,s)$ then also vanishes for $x \leq 0$ and $s$ in a suitable interval of the real axis. The "two-sided" Laplace transform of $\widehat{w}_+(x,t)$ is then no different from a one-sided transform, that is,

$$W_+(\xi, s) = \int_0^\infty \widehat{w}_+(x,s)\, e^{-\xi x}\, dx\,. \tag{1.3.16}$$

Evidently, the value of $W_+(\xi, s)$ at any particular value of $\xi$ depends on $\widehat{w}_+$ over the entire range of $x$. Likewise, from the inversion integral, the value of $\widehat{w}_+(x,s)$ at any $x$ depends on $W_+$ over the full range of $\xi$. However, transform pairs of the kind defined by (1.3.16) have remarkable asymptotic properties relating limiting values of a function and its transform. According to van der Pol and Bremmer (1955), theorems that yield an asymptotic property of a transform from a known asymptotic property of the function are called Abel theorems. On the other hand, theorems that provide an asymptotic property of a function from a known asymptotic property of its transform are called Tauber theorems.

The basic Abel theorem has been useful in elastic fracture mechanics studies. In this case, direct asymptotic analysis of the differential equations often leads to the conclusion that a particular field depends on a spatial coordinate near the crack tip in a certain way. Through an Abel theorem, it is then possible to make definite statements about the behavior of the transform at remote points in the transform parameter plane.

For example, suppose that $\widehat{w}_+(x,s)$ in (1.3.16) has the behavior

$$\widehat{w}_+(x,s) \sim x^\alpha \tag{1.3.17}$$

near $x = 0$, where $\alpha$ is a real constant with $\alpha > -1$. The Abel theorem then states that

$$\lim_{x \to 0^+} \frac{\Gamma(1+\alpha)}{x^\alpha} \widehat{w}_+(x,s) = \lim_{\xi \to +\infty} \xi^{1+\alpha} W_+(\xi, s)\,, \tag{1.3.18}$$

where $\Gamma(\cdot)$ is the Gamma or factorial function. If the region of convergence of the integral defining $W_+$ includes the imaginary axis

then it can also be established that

$$\lim_{x\to+\infty} \frac{\Gamma(1+\beta)}{x^{\beta}} \widehat{w}_+(x,s) = \lim_{\xi\to 0^+} \xi^{1+\beta} W_+(\xi,s) \tag{1.3.19}$$

for an appropriate value of $\beta > -1$.

## 1.4 Overview of dynamic fracture mechanics

Analytical results of a permanent nature in the area of dynamic fracture mechanics are gathered together and summarized in Chapters 2 through 8. For the most part, the results are obtained within the small strain formulation of continuum mechanics as described in Section 1.2.2. Cracks are assumed to be planar and the opposite faces are assumed to occupy the same plane in the undeformed configuration. The geometrical configurations considered are also very simple, in general. For example, numerous developments are based on the assumption of a half plane crack in an otherwise unbounded body under conditions that result in a two-dimensional state of deformation. Though this assumption provides the basis for the formulation of a well-posed boundary value problem, it does have practical limitations. The results can only be applied in cases when the material points of interest are much closer to the crack edge than to other boundaries of the body and when the local deformation field is indeed essentially two-dimensional. Other configurations considered include crack growth in a long uniform strip and idealizations of bodies as structural elements, typically beams or strings.

The purpose here is to provide a very brief overview of the material in the chapters to follow, including a summary of the main results and comments on directions for further study. A number of books and monographs are available to provide background on fracture mechanics from various perspectives. For example, the books by Kanninen and Popelar (1985), Cherepanov (1979), and Hellan (1984) develop the topic from the mechanics perspective. The books by Knott (1979) and Lawn and Wilshaw (1975) introduce the subject from a materials science perspective, whereas Barsom and Rolfe (1987) discuss fracture within the context of materials selection and structural applications.

A number of interesting but specialized issues in dynamic fracture are discussed only briefly in this monograph. Among these topics are: the existence of an apparent terminal crack speed well below

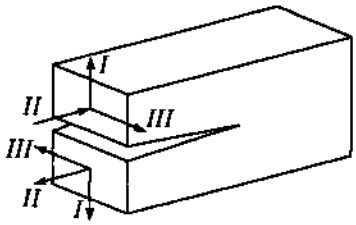


Figure 1.4. The classification of crack opening modes I, II, and III based on the components of the relative displacement of the crack faces near the crack edge.

the Rayleigh wave speed in glass and some other very brittle materials; the contact interaction of crack faces during dynamic fracture events in materials; the symmetric bifurcation of a straight, rapidly running crack in a tensile field into two branches; the influence of heat generated during rapid fracture on the separation process or on the stress distribution; atomistic modeling of dynamic cleavage; and the erosion of materials by impact of solid particles or liquid droplets. Some experimental research and analytical modeling have been done on each of these problems using the approaches described in the following chapters, but the phenomena are not yet completely understood. Available references are included in the Bibliography. Other topical areas that are not yet fully developed are identified in the overview to follow or in the later chapters.

### *1.4.1 Basic elastodynamic solutions for a stationary crack*

In Chapter 2, the fundamental stationary crack problems of linear elastodynamics are analyzed. The two-dimensional configuration of a semi-infinite crack in an otherwise unbounded body is considered. The body is initially stress free and at rest. At time $t = 0$, equal and opposite spatially uniform tractions begin to act on the crack faces. Formulated in this way, the problem has neither a characteristic length nor a characteristic time. The in-plane opening mode of crack deformation (mode I), the in-plane shearing mode (mode II), and the antiplane shearing mode (mode III) are treated separately. The direction of the relative crack opening displacement for each mode is indicated in Figure 1.4, and the classification is further discussed in Section 2.1.

Two mathematical methods for solving initial-boundary value problems are introduced, namely, Green's method based on superposition over a fundamental singular solution of the governing differential

equation and the Wiener-Hopf method based on Laplace transform techniques. These methods are presented in a formal way, and they are used in later chapters as well. The main results of the analysis are very simple. Indeed, aside from a scalar multiplier of order unity, the stress distribution near the crack edge can be obtained for each mode through dimensional analysis. With reference to the coordinate system shown in Figure 1.4, if the crack face traction has magnitude $\tau^*$ per unit area and if it acts in a direction parallel to the crack edge, then the state of deformation is antiplane shear and the crack opens in mode III. The shear stress on the plane $y = 0$ directly ahead of the crack tip, and close to it in some sense, is

$$\sigma_{zy}(x,0,t) \sim \frac{K_{III}(t)}{\sqrt{2\pi x}}, \qquad K_{III}(t) = 2\tau^* \sqrt{\frac{2c_s t}{\pi}},$$

where $c_s$ is the elastic shear wave speed. This result is derived in Section 2.3 where it appears as (2.3.18). Likewise, if a normal pressure of magnitude $\sigma^*$ begins to act on the crack faces at time $t = 0$ then the state of deformation is plane strain and the crack opens in mode I. The normal stress on the plane $y = 0$ near the crack tip in this case is

$$\sigma_{yy}(x,0,t) \sim \frac{K_I(t)}{\sqrt{2\pi x}}, \qquad K_I(t) = 2\sigma^* \frac{\sqrt{c_d t(1-2\nu)/\pi}}{(1-\nu)},$$

where $c_d$ is the dilatational wave speed of the material and $\nu$ is Poisson's ratio. This result is derived in Section 2.5.3 where it appears as (2.5.44). Lastly, if a shear traction of magnitude $\tau^*$ begins to act on the crack faces at time $t = 0$ in a direction perpendicular to the crack edge, then the state of deformation is plane strain and the crack opens in mode II. The shear stress on the plane directly ahead of the crack tip in this case is

$$\sigma_{xy}(x,0,t) \sim \frac{K_{II}(t)}{\sqrt{2\pi x}}, \qquad K_{II}(t) = 2\tau^* \sqrt{\frac{2c_s t}{\pi(1-\nu)}}.$$

This result is given in Section 2.6 as equation (2.6.9). The solutions obtained in Chapter 2 are readily generalized to the case of applied crack face loading with arbitrary time variation. Furthermore, the solution for mode I, II, or III is identical to that for a plane tensile, transverse shear, or horizontal shear wave carrying a simple jump